```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
Seeds: [42, 52, 62, 72, 82]

Checking dataset paths...
Dataset found at:
/content/drive/MyDrive/Bone_Fracture_Binary_Classification

================ DATASET INFORMATION ================
Training images   : 9246
Validation images : 829
Testing images    : 506
Classes           : ['fractured', 'not fractured']
Class to index    : {'fractured': 0, 'not fractured': 1}

Using Device: cuda
GPU: Tesla T4

================ MODEL ARCHITECTURE ================
Downloading: "https://download.pytorch.org/models/efficientnet_b0_rwightman-7f5810bc.pth" to /root/.cache/torch/hub/checkpoints/efficientnet_b0_
100%|██████████| 20.5M/20.5M [00:00<00:00, 175MB/s]
EfficientNet_16(
  (backbone): EfficientNet(
    (features): Sequential(
      (0): Conv2dNormActivation(
        (0): Conv2d(3, 32, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bias=False)
        (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (2): SiLU(inplace=True)
      )
      (1): Sequential(
        (0): MBConv(
          (block): Sequential(
            (0): Conv2dNormActivation(
              (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=32, bias=False)
              (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
              (2): SiLU(inplace=True)
            )
            (1): SqueezeExcitation(
              (avgpool): AdaptiveAvgPool2d(output_size=1)
              (fc1): Conv2d(32, 8, kernel_size=(1, 1), stride=(1, 1))
              (fc2): Conv2d(8, 32, kernel_size=(1, 1), stride=(1, 1))
              (activation): SiLU(inplace=True)
              (scale_activation): Sigmoid()
            )
            (2): Conv2dNormActivation(
              (0): Conv2d(32, 16, kernel_size=(1, 1), stride=(1, 1), bias=False)
              (1): BatchNorm2d(16, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            )
          )
          (stochastic_depth): StochasticDepth(p=0.0, mode=row)
        )
      )
      (2): Sequential(
        (0): MBConv(
          (block): Sequential(
            (0): Conv2dNormActivation(
              (0): Conv2d(16, 96, kernel_size=(1, 1), stride=(1, 1), bias=False)
              (1): BatchNorm2d(96, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
              (2): SiLU(inplace=True)
            )
            (1): Conv2dNormActivation(
              (0): Conv2d(96, 96, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), groups=96, bias=False)
              (1): BatchNorm2d(96, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
              (2): SiLU(inplace=True)
            )
            (2): SqueezeExcitation(
              (avgpool): AdaptiveAvgPool2d(output_size=1)
              (fc1): Conv2d(96, 4, kernel_size=(1, 1), stride=(1, 1))
              (fc2): Conv2d(4, 96, kernel_size=(1, 1), stride=(1, 1))
              (activation): SiLU(inplace=True)
              (scale_activation): Sigmoid()
            )
            (3): Conv2dNormActivation(
              (0): Conv2d(96, 24, kernel_size=(1, 1), stride=(1, 1), bias=False)
              (1): BatchNorm2d(24, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            )
          )
          (stochastic_depth): StochasticDepth(p=0.0125, mode=row)
        )
        (1): MBConv(
          (block): Sequential(
            (0): Conv2dNormActivation(
              (0): Conv2d(24, 144, kernel_size=(1, 1), stride=(1, 1), bias=False)
              (1): BatchNorm2d(144, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
              (2): SiLU(inplace=True)
            )
            (1): Conv2dNormActivation(
```

```
          (0): Conv2d(144, 144, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=144, bias=False)
          (1): BatchNorm2d(144, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(144, 6, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(6, 144, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(144, 24, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(24, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.025, mode=row)
    )
  )
  (3): Sequential(
    (0): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(24, 144, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(144, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(144, 144, kernel_size=(5, 5), stride=(2, 2), padding=(2, 2), groups=144, bias=False)
          (1): BatchNorm2d(144, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(144, 6, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(6, 144, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(144, 40, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(40, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.037500000000000006, mode=row)
    )
    (1): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(40, 240, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(240, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(240, 240, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=240, bias=False)
          (1): BatchNorm2d(240, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(240, 10, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(10, 240, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(240, 40, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(40, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.05, mode=row)
    )
  )
  (4): Sequential(
    (0): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(40, 240, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(240, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(240, 240, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), groups=240, bias=False)
          (1): BatchNorm2d(240, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
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        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(240, 10, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(10, 240, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(240, 80, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(80, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.0625, mode=row)
    )
    (1): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(80, 480, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(480, 480, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=480, bias=False)
          (1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(480, 20, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(20, 480, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(480, 80, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(80, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.07500000000000001, mode=row)
    )
    (2): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(80, 480, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(480, 480, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=480, bias=False)
          (1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(480, 20, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(20, 480, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(480, 80, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(80, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.08750000000000001, mode=row)
    )
  )
  (5): Sequential(
    (0): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(80, 480, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(480, 480, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=480, bias=False)
          (1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(480, 20, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(20, 480, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
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          )
          (3): Conv2dNormActivation(
            (0): Conv2d(480, 112, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(112, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
        (stochastic_depth): StochasticDepth(p=0.1, mode=row)
      )
      (1): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(112, 672, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (1): Conv2dNormActivation(
            (0): Conv2d(672, 672, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=672, bias=False)
            (1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(672, 28, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(28, 672, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(672, 112, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(112, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
        (stochastic_depth): StochasticDepth(p=0.1125, mode=row)
      )
      (2): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(112, 672, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (1): Conv2dNormActivation(
            (0): Conv2d(672, 672, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=672, bias=False)
            (1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(672, 28, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(28, 672, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(672, 112, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(112, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
        (stochastic_depth): StochasticDepth(p=0.125, mode=row)
      )
    )
    (6): Sequential(
      (0): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(112, 672, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (1): Conv2dNormActivation(
            (0): Conv2d(672, 672, kernel_size=(5, 5), stride=(2, 2), padding=(2, 2), groups=672, bias=False)
            (1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(672, 28, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(28, 672, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(672, 192, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
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      (stochastic_depth): StochasticDepth(p=0.1375, mode=row)
    )
    (1): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(192, 1152, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(1152, 1152, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=1152, bias=False)
          (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(1152, 48, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(48, 1152, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(1152, 192, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.15000000000000002, mode=row)
    )
    (2): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(192, 1152, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(1152, 1152, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=1152, bias=False)
          (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(1152, 48, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(48, 1152, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(1152, 192, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.1625, mode=row)
    )
    (3): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(192, 1152, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(1152, 1152, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=1152, bias=False)
          (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(1152, 48, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(48, 1152, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(1152, 192, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.17500000000000002, mode=row)
    )
  )
  (7): Sequential(
    (0): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(192, 1152, kernel_size=(1, 1), stride=(1, 1), bias=False)
```

```
          (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(1152, 1152, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=1152, bias=False)
          (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(1152, 48, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(48, 1152, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(1152, 320, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(320, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.1875, mode=row)
    )
  )
  (8): Conv2dNormActivation(
    (0): Conv2d(320, 1280, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (1): BatchNorm2d(1280, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): SiLU(inplace=True)
  )
 )
 (avgpool): AdaptiveAvgPool2d(output_size=1)
 (classifier): Identity()
)
(fc1): Linear(in_features=1280, out_features=16, bias=True)
(tanh): Tanh()
(fc2): Linear(in_features=16, out_features=1, bias=True)
)


================================================================================
STARTING TRAINING FOR SEED 42
================================================================================

Random Seed Set To: 42
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 42, Epoch 1

Seed 42 | Epoch 1/10 | Train Loss: 0.3167 | Val Loss: 0.1715 | Train Acc: 89.54% | Val Acc: 95.66%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 42, Epoch 2

Seed 42 | Epoch 2/10 | Train Loss: 0.0995 | Val Loss: 0.1331 | Train Acc: 99.06% | Val Acc: 97.10%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 42, Epoch 3

Seed 42 | Epoch 3/10 | Train Loss: 0.0757 | Val Loss: 0.1091 | Train Acc: 99.39% | Val Acc: 97.23%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 42, Epoch 4

Seed 42 | Epoch 4/10 | Train Loss: 0.0680 | Val Loss: 0.0967 | Train Acc: 99.36% | Val Acc: 98.19%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 42 | Epoch 5/10 | Train Loss: 0.0615 | Val Loss: 0.1014 | Train Acc: 99.68% | Val Acc: 97.71%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 42 | Epoch 6/10 | Train Loss: 0.0606 | Val Loss: 0.1004 | Train Acc: 99.69% | Val Acc: 97.83%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 42 | Epoch 7/10 | Train Loss: 0.0594 | Val Loss: 0.1010 | Train Acc: 99.72% | Val Acc: 97.59%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 42 | Epoch 8/10 | Train Loss: 0.0589 | Val Loss: 0.1017 | Train Acc: 99.73% | Val Acc: 97.83%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 42 | Epoch 9/10 | Train Loss: 0.0597 | Val Loss: 0.1000 | Train Acc: 99.62% | Val Acc: 97.83%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 42 | Epoch 10/10 | Train Loss: 0.0596 | Val Loss: 0.1015 | Train Acc: 99.66% | Val Acc: 97.95%


================================================================================
SEED 42 TRAINING COMPLETE
Time taken: 42.11 minutes
```

```
Best Validation Loss: 0.096745
Model saved: /content/best_model_seed_42.pth
================================================================================

Evaluating Seed 42 on Training / Validation / Testing...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(

==========================================================================================
SEED 42 PERFORMANCE
==========================================================================================
         Metric  Training  Validation   Testing
   Accuracy (%) 99.848583   98.190591 97.035573
Sensitivity (%) 99.870690   98.373984 97.014925
Specificity (%) 99.826314   97.922849 97.058824
  Precision (%) 99.827660   98.574338 97.378277
   F1-Score (%) 99.849170   98.474059 97.196262
 Error Rate (%)  0.151417    1.809409  2.964427
        ROC-AUC  0.999975    0.997835  0.994670


================================================================================
STARTING TRAINING FOR SEED 52
================================================================================

Random Seed Set To: 52
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 1

Seed 52 | Epoch 1/10 | Train Loss: 0.3624 | Val Loss: 0.2187 | Train Acc: 89.31% | Val Acc: 94.93%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 2

Seed 52 | Epoch 2/10 | Train Loss: 0.1511 | Val Loss: 0.1688 | Train Acc: 98.96% | Val Acc: 96.50%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 3

Seed 52 | Epoch 3/10 | Train Loss: 0.1169 | Val Loss: 0.1228 | Train Acc: 99.37% | Val Acc: 98.43%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 4

Seed 52 | Epoch 4/10 | Train Loss: 0.1010 | Val Loss: 0.1208 | Train Acc: 99.66% | Val Acc: 98.31%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 5

Seed 52 | Epoch 5/10 | Train Loss: 0.0981 | Val Loss: 0.1163 | Train Acc: 99.69% | Val Acc: 98.67%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 6

Seed 52 | Epoch 6/10 | Train Loss: 0.0969 | Val Loss: 0.1149 | Train Acc: 99.65% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 7

Seed 52 | Epoch 7/10 | Train Loss: 0.0939 | Val Loss: 0.1125 | Train Acc: 99.75% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 52 | Epoch 8/10 | Train Loss: 0.0939 | Val Loss: 0.1147 | Train Acc: 99.74% | Val Acc: 98.67%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 9

Seed 52 | Epoch 9/10 | Train Loss: 0.0942 | Val Loss: 0.1103 | Train Acc: 99.74% | Val Acc: 99.03%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 52 | Epoch 10/10 | Train Loss: 0.0940 | Val Loss: 0.1139 | Train Acc: 99.70% | Val Acc: 98.79%


================================================================================
SEED 52 TRAINING COMPLETE
Time taken: 8.39 minutes
Best Validation Loss: 0.110307
Model saved: /content/best_model_seed_52.pth
================================================================================

Evaluating Seed 52 on Training / Validation / Testing...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(

==========================================================================================
```

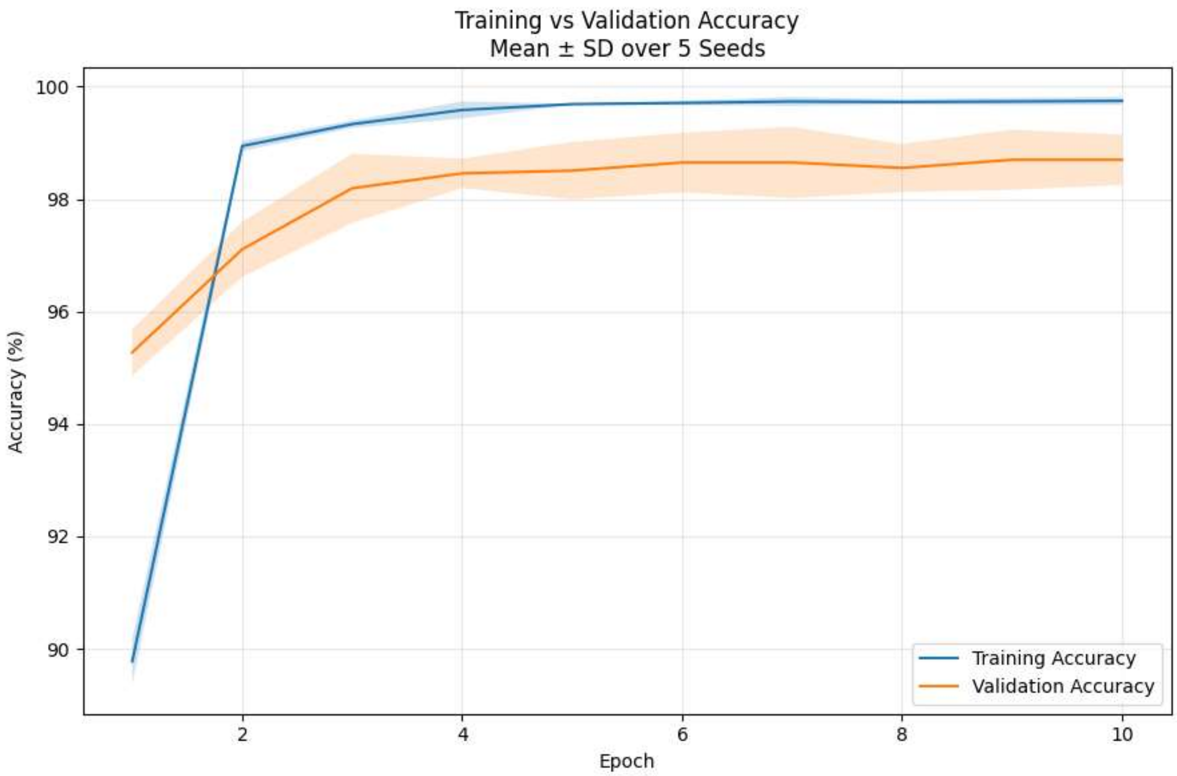
```
SEED 52 PERFORMANCE
==========================================================================================
         Metric  Training  Validation   Testing
   Accuracy (%) 99.924292   99.034982 98.616601
Sensitivity (%) 99.935345   99.390244 98.880597
Specificity (%) 99.913157   98.516320 98.319328
  Precision (%) 99.913812   98.987854 98.513011
   F1-Score (%) 99.924577   99.188641 98.696462
 Error Rate (%)  0.075708    0.965018  1.383399
        ROC-AUC  0.999995    0.998884  0.995359


================================================================================
STARTING TRAINING FOR SEED 62
================================================================================

Random Seed Set To: 62
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 1

Seed 62 | Epoch 1/10 | Train Loss: 0.3259 | Val Loss: 0.1899 | Train Acc: 89.75% | Val Acc: 95.78%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 2

Seed 62 | Epoch 2/10 | Train Loss: 0.1220 | Val Loss: 0.1440 | Train Acc: 98.80% | Val Acc: 96.86%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 3

Seed 62 | Epoch 3/10 | Train Loss: 0.0941 | Val Loss: 0.1119 | Train Acc: 99.28% | Val Acc: 97.95%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 4

Seed 62 | Epoch 4/10 | Train Loss: 0.0795 | Val Loss: 0.1054 | Train Acc: 99.49% | Val Acc: 98.31%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 5

Seed 62 | Epoch 5/10 | Train Loss: 0.0752 | Val Loss: 0.1018 | Train Acc: 99.70% | Val Acc: 98.31%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 6

Seed 62 | Epoch 6/10 | Train Loss: 0.0733 | Val Loss: 0.1007 | Train Acc: 99.74% | Val Acc: 98.43%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 7

Seed 62 | Epoch 7/10 | Train Loss: 0.0740 | Val Loss: 0.0995 | Train Acc: 99.61% | Val Acc: 98.55%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 62 | Epoch 8/10 | Train Loss: 0.0729 | Val Loss: 0.0997 | Train Acc: 99.64% | Val Acc: 98.55%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 9

Seed 62 | Epoch 9/10 | Train Loss: 0.0718 | Val Loss: 0.0988 | Train Acc: 99.76% | Val Acc: 98.55%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 10

Seed 62 | Epoch 10/10 | Train Loss: 0.0721 | Val Loss: 0.0986 | Train Acc: 99.74% | Val Acc: 98.79%


================================================================================
SEED 62 TRAINING COMPLETE
Time taken: 8.26 minutes
Best Validation Loss: 0.098574
Model saved: /content/best_model_seed_62.pth
================================================================================

Evaluating Seed 62 on Training / Validation / Testing...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(

==========================================================================================
SEED 62 PERFORMANCE
==========================================================================================
         Metric  Training  Validation   Testing
   Accuracy (%) 99.837768   98.793727 98.023715
Sensitivity (%) 99.849138   98.780488 97.761194
Specificity (%) 99.826314   98.813056 98.319328
  Precision (%) 99.827623   99.183673 98.496241
```

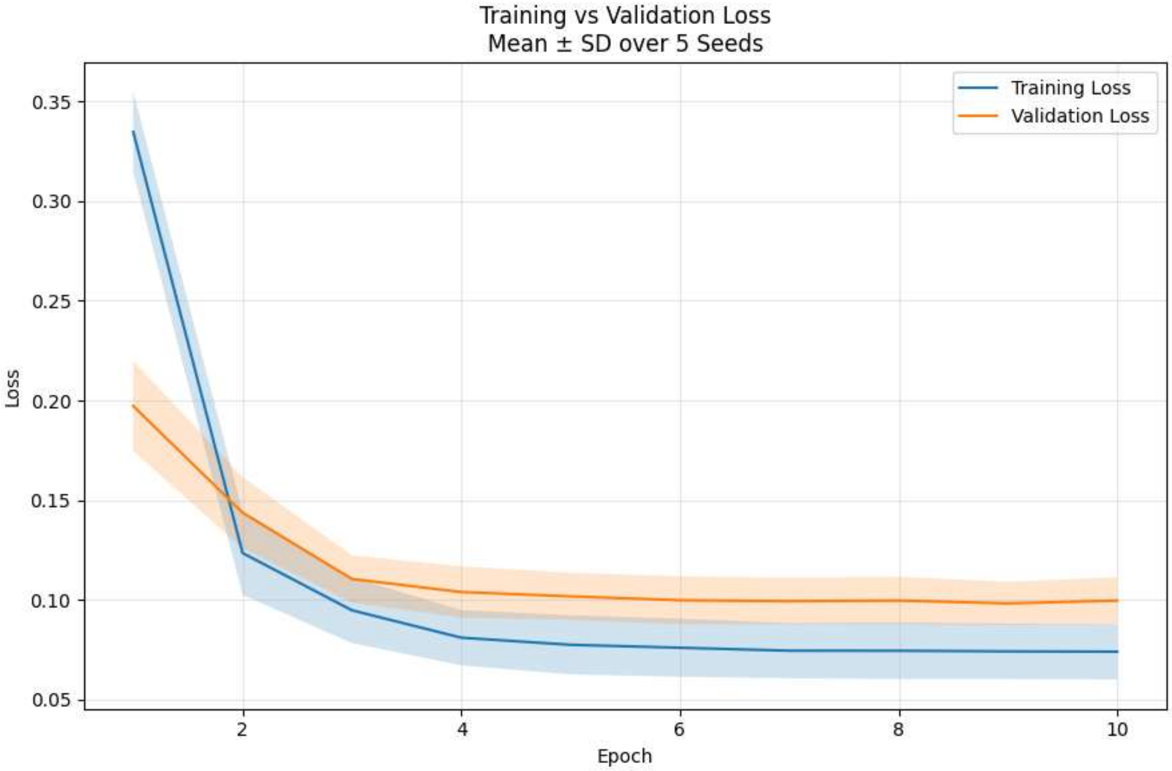
```
    F1-Score (%) 99.838379   98.981670 98.127341
  Error Rate (%)  0.162232    1.206273  1.976285
         ROC-AUC  0.999986    0.998975  0.996535


================================================================================
STARTING TRAINING FOR SEED 72
================================================================================

Random Seed Set To: 72
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 1

Seed 72 | Epoch 1/10 | Train Loss: 0.3189 | Val Loss: 0.1833 | Train Acc: 90.41% | Val Acc: 94.93%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 2

Seed 72 | Epoch 2/10 | Train Loss: 0.1083 | Val Loss: 0.1218 | Train Acc: 98.95% | Val Acc: 97.83%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 3

Seed 72 | Epoch 3/10 | Train Loss: 0.0833 | Val Loss: 0.0916 | Train Acc: 99.24% | Val Acc: 98.67%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 4

Seed 72 | Epoch 4/10 | Train Loss: 0.0690 | Val Loss: 0.0870 | Train Acc: 99.66% | Val Acc: 98.67%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 5

Seed 72 | Epoch 5/10 | Train Loss: 0.0668 | Val Loss: 0.0836 | Train Acc: 99.69% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 6

Seed 72 | Epoch 6/10 | Train Loss: 0.0657 | Val Loss: 0.0814 | Train Acc: 99.74% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 7

Seed 72 | Epoch 7/10 | Train Loss: 0.0641 | Val Loss: 0.0807 | Train Acc: 99.75% | Val Acc: 99.03%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 72 | Epoch 8/10 | Train Loss: 0.0641 | Val Loss: 0.0815 | Train Acc: 99.75% | Val Acc: 98.79%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 72 | Epoch 9/10 | Train Loss: 0.0640 | Val Loss: 0.0807 | Train Acc: 99.74% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 72 | Epoch 10/10 | Train Loss: 0.0635 | Val Loss: 0.0816 | Train Acc: 99.77% | Val Acc: 98.79%


================================================================================
SEED 72 TRAINING COMPLETE
Time taken: 8.38 minutes
Best Validation Loss: 0.080658
Model saved: /content/best_model_seed_72.pth
================================================================================

Evaluating Seed 72 on Training / Validation / Testing...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(

==========================================================================================
SEED 72 PERFORMANCE
==========================================================================================
          Metric  Training  Validation   Testing
    Accuracy (%) 99.945923   99.034982 98.814229
 Sensitivity (%) 99.956897   99.593496 99.253731
 Specificity (%) 99.934868   98.219585 98.319328
   Precision (%) 99.935359   98.790323 98.518519
    F1-Score (%) 99.946126   99.190283 98.884758
  Error Rate (%)  0.054077    0.965018  1.185771
         ROC-AUC  0.999997    0.999608  0.999655


================================================================================
STARTING TRAINING FOR SEED 82
================================================================================

Random Seed Set To: 82
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
```

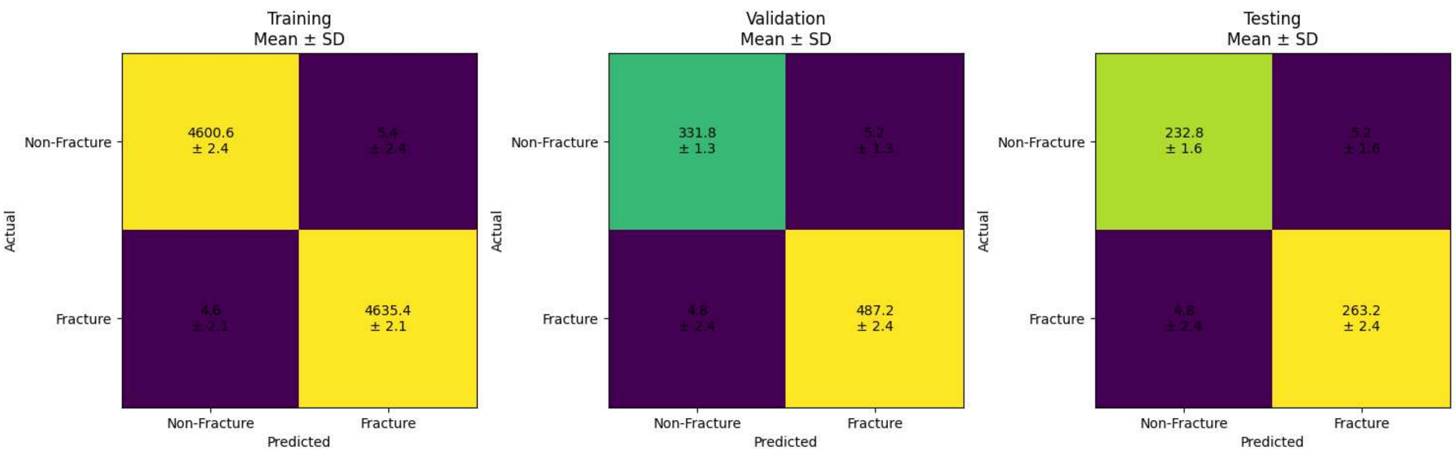
```
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 1

Seed 82 | Epoch 1/10 | Train Loss: 0.3490 | Val Loss: 0.2229 | Train Acc: 89.89% | Val Acc: 95.05%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 2

Seed 82 | Epoch 2/10 | Train Loss: 0.1370 | Val Loss: 0.1512 | Train Acc: 98.94% | Val Acc: 97.23%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 3

Seed 82 | Epoch 3/10 | Train Loss: 0.1043 | Val Loss: 0.1174 | Train Acc: 99.37% | Val Acc: 98.67%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 4

Seed 82 | Epoch 4/10 | Train Loss: 0.0882 | Val Loss: 0.1101 | Train Acc: 99.73% | Val Acc: 98.79%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 5

Seed 82 | Epoch 5/10 | Train Loss: 0.0863 | Val Loss: 0.1062 | Train Acc: 99.69% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 6

Seed 82 | Epoch 6/10 | Train Loss: 0.0839 | Val Loss: 0.1022 | Train Acc: 99.72% | Val Acc: 99.16%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 82 | Epoch 7/10 | Train Loss: 0.0818 | Val Loss: 0.1035 | Train Acc: 99.84% | Val Acc: 99.16%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 8

Seed 82 | Epoch 8/10 | Train Loss: 0.0831 | Val Loss: 0.1011 | Train Acc: 99.76% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 82 | Epoch 9/10 | Train Loss: 0.0815 | Val Loss: 0.1018 | Train Acc: 99.81% | Val Acc: 99.16%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 82 | Epoch 10/10 | Train Loss: 0.0814 | Val Loss: 0.1031 | Train Acc: 99.85% | Val Acc: 99.16%


================================================================================
SEED 82 TRAINING COMPLETE
Time taken: 8.43 minutes
Best Validation Loss: 0.101117
Model saved: /content/best_model_seed_82.pth
================================================================================

Evaluating Seed 82 on Training / Validation / Testing...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(

==========================================================================================
SEED 82 PERFORMANCE
==========================================================================================
         Metric  Training  Validation   Testing
   Accuracy (%) 99.902661   98.914355 97.628458
Sensitivity (%) 99.892241   98.983740 98.134328
Specificity (%) 99.913157   98.813056 97.058824
  Precision (%) 99.913775   99.185336 97.407407
   F1-Score (%) 99.903007   99.084435 97.769517
 Error Rate (%)  0.097339    1.085645  2.371542
        ROC-AUC  0.999801    0.998709  0.997664


==========================================================================================
ALL FIVE SEEDS COMPLETED
==========================================================================================
Seeds completed: [42, 52, 62, 72, 82]
Total time: 82.92 minutes


==========================================================================================
PER-SEED PERFORMANCE RESULTS
==========================================================================================
 Seed    Dataset          Metric     Value
   42   Training    Accuracy (%) 99.848583
   42   Training Sensitivity (%) 99.870690
   42   Training Specificity (%) 99.826314
   42   Training   Precision (%) 99.827660
   42   Training    F1-Score (%) 99.849170
   42   Training  Error Rate (%)  0.151417
```

```
    42   Training         ROC-AUC  0.999975
    42 Validation    Accuracy (%) 98.190591
    42 Validation Sensitivity (%) 98.373984
    42 Validation Specificity (%) 97.922849
    42 Validation   Precision (%) 98.574338
    42 Validation    F1-Score (%) 98.474059
    42 Validation  Error Rate (%)  1.809409
    42 Validation         ROC-AUC  0.997835
    42    Testing    Accuracy (%) 97.035573
    42    Testing Sensitivity (%) 97.014925
    42    Testing Specificity (%) 97.058824
    42    Testing   Precision (%) 97.378277
    42    Testing    F1-Score (%) 97.196262
    42    Testing  Error Rate (%)  2.964427
    42    Testing         ROC-AUC  0.994670
    52   Training    Accuracy (%) 99.924292
    52   Training Sensitivity (%) 99.935345
    52   Training Specificity (%) 99.913157
    52   Training   Precision (%) 99.913812
    52   Training    F1-Score (%) 99.924577
    52   Training  Error Rate (%)  0.075708
    52   Training         ROC-AUC  0.999995
    52 Validation    Accuracy (%) 99.034982
    52 Validation Sensitivity (%) 99.390244
    52 Validation Specificity (%) 98.516320
    52 Validation   Precision (%) 98.987854
    52 Validation    F1-Score (%) 99.188641
    52 Validation  Error Rate (%)  0.965018
    52 Validation         ROC-AUC  0.998884
    52    Testing    Accuracy (%) 98.616601
    52    Testing Sensitivity (%) 98.880597
    52    Testing Specificity (%) 98.319328
    52    Testing   Precision (%) 98.513011
    52    Testing    F1-Score (%) 98.696462
    52    Testing  Error Rate (%)  1.383399
    52    Testing         ROC-AUC  0.995359
    62   Training    Accuracy (%) 99.837768
    62   Training Sensitivity (%) 99.849138
    62   Training Specificity (%) 99.826314
    62   Training   Precision (%) 99.827623
    62   Training    F1-Score (%) 99.838379
    62   Training  Error Rate (%)  0.162232
    62   Training         ROC-AUC  0.999986
    62 Validation    Accuracy (%) 98.793727
    62 Validation Sensitivity (%) 98.780488
    62 Validation Specificity (%) 98.813056
    62 Validation   Precision (%) 99.183673
    62 Validation    F1-Score (%) 98.981670
    62 Validation  Error Rate (%)  1.206273
    62 Validation         ROC-AUC  0.998975
    62    Testing    Accuracy (%) 98.023715
    62    Testing Sensitivity (%) 97.761194
    62    Testing Specificity (%) 98.319328
    62    Testing   Precision (%) 98.496241
    62    Testing    F1-Score (%) 98.127341
    62    Testing  Error Rate (%)  1.976285
    62    Testing         ROC-AUC  0.996535
    72   Training    Accuracy (%) 99.945923
    72   Training Sensitivity (%) 99.956897
    72   Training Specificity (%) 99.934868
    72   Training   Precision (%) 99.935359
    72   Training    F1-Score (%) 99.946126
    72   Training  Error Rate (%)  0.054077
    72   Training         ROC-AUC  0.999997
    72 Validation    Accuracy (%) 99.034982
    72 Validation Sensitivity (%) 99.593496
    72 Validation Specificity (%) 98.219585
    72 Validation   Precision (%) 98.790323
    72 Validation    F1-Score (%) 99.190283
    72 Validation  Error Rate (%)  0.965018
    72 Validation         ROC-AUC  0.999608
    72    Testing    Accuracy (%) 98.814229
    72    Testing Sensitivity (%) 99.253731
    72    Testing Specificity (%) 98.319328
    72    Testing   Precision (%) 98.518519
    72    Testing    F1-Score (%) 98.884758
    72    Testing  Error Rate (%)  1.185771
    72    Testing         ROC-AUC  0.999655
    82   Training    Accuracy (%) 99.902661
    82   Training Sensitivity (%) 99.892241
    82   Training Specificity (%) 99.913157
    82   Training   Precision (%) 99.913775
    82   Training    F1-Score (%) 99.903007
    82   Training  Error Rate (%)  0.097339
    82   Training         ROC-AUC  0.999801
    82 Validation    Accuracy (%) 98.914355
```

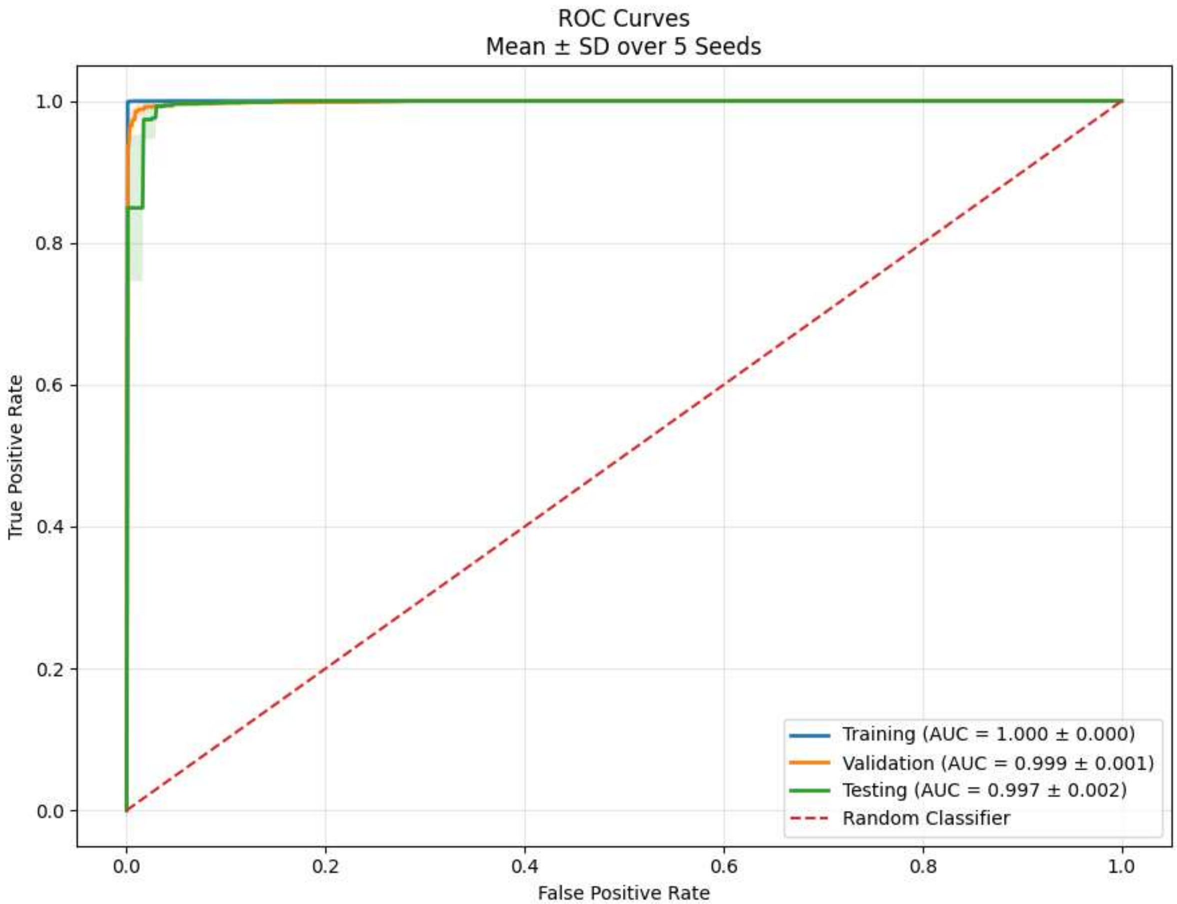
```
    82 Validation Sensitivity (%) 98.983740
    82 Validation Specificity (%) 98.813056
    82 Validation   Precision (%) 99.185336
    82 Validation    F1-Score (%) 99.084435
    82 Validation  Error Rate (%)  1.085645
    82 Validation          ROC-AUC  0.998709
    82    Testing    Accuracy (%) 97.628458
    82    Testing Sensitivity (%) 98.134328
    82    Testing Specificity (%) 97.058824
    82    Testing   Precision (%) 97.407407
    82    Testing    F1-Score (%) 97.769517
    82    Testing  Error Rate (%)  2.371542
    82    Testing          ROC-AUC  0.997664


==========================================================================================
FINAL PERFORMANCE COMPARISON TABLE
MEAN ± SD OVER FIVE SEEDS
==========================================================================================
         Metric     Training   Validation      Testing
   Accuracy (%) 99.89 ± 0.05 98.79 ± 0.35 98.02 ± 0.73
Sensitivity (%) 99.90 ± 0.04 99.02 ± 0.49 98.21 ± 0.89
Specificity (%) 99.88 ± 0.05 98.46 ± 0.39 97.82 ± 0.69
  Precision (%) 99.88 ± 0.05 98.94 ± 0.26 98.06 ± 0.61
   F1-Score (%) 99.89 ± 0.05 98.98 ± 0.30 98.13 ± 0.69
 Error Rate (%)  0.11 ± 0.05  1.21 ± 0.35  1.98 ± 0.73
        ROC-AUC  1.00 ± 0.00  1.00 ± 0.00  1.00 ± 0.00


==========================================================================================
TIME TAKEN FOR EACH SEED
==========================================================================================
 Seed  Time (minutes)
   42     2526.628683
   52      503.682524
   62      495.797784
   72      502.635289
   82      505.559842

Total execution time: 82.92 minutes
```








```
======================================================================================
MEAN ± SD CONFUSION MATRICES
======================================================================================

Training Confusion Matrix:
[[4.60e+03 5.40e+00]
 [4.60e+00 4.64e+03]]

Training CM SD:
[[2.41 2.41]
 [2.07 2.07]]

Validation Confusion Matrix:
[[331.8   5.2]
 [  4.8 487.2]]

Validation CM SD:
[[1.3  1.3 ]
 [2.39 2.39]]

Testing Confusion Matrix:
[[232.8   5.2]
 [  4.8 263.2]]

Testing CM SD:
[[1.64 1.64]
 [2.39 2.39]]
```




```
==========================================================================================
ROC-AUC MEAN ± SD
==========================================================================================
Training   : 1.0000 ± 0.0001
Validation : 0.9988 ± 0.0006
Testing    : 0.9968 ± 0.0020


====================================================================================================
PERFORMANCE COMPARISON TABLE
MEAN ± SD OVER FIVE SEEDS
====================================================================================================
         Metric     Training   Validation      Testing
   Accuracy (%) 99.89 ± 0.05 98.79 ± 0.35 98.02 ± 0.73
Sensitivity (%) 99.90 ± 0.04 99.02 ± 0.49 98.21 ± 0.89
Specificity (%) 99.88 ± 0.05 98.46 ± 0.39 97.82 ± 0.69
  Precision (%) 99.88 ± 0.05 98.94 ± 0.26 98.06 ± 0.61
   F1-Score (%) 99.89 ± 0.05 98.98 ± 0.30 98.13 ± 0.69
 Error Rate (%)  0.11 ± 0.05  1.21 ± 0.35  1.98 ± 0.73
        ROC-AUC  1.00 ± 0.00  1.00 ± 0.00  1.00 ± 0.00


==========================================================================================
RESULT FILES SAVED
==========================================================================================
/content/final_mean_sd_results.csv
/content/per_seed_results.csv
/content/seed_time_results.csv


====================================================================================================
FINAL EXPERIMENT SUMMARY
====================================================================================================
Architecture:
EfficientNet-B0 → 1280 → Linear(1280→16) → Tanh → Linear(16→1) → Sigmoid

Seeds: [42, 52, 62, 72, 82]

Epochs per seed: 10

Batch size: 64

Learning rate: 0.0001
```

```
Device: cuda

Total execution time: 82.92 minutes

All processing completed successfully.
```